



OFFICIAL

Deep Vein Thrombosis (DVT) Pathway Guideline – Emergency Department

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1. Purpose

This pathway outlines the assessment of patients with a suspected DVT, treatment options for a confirmed DVT and tests required prior to commencing anticoagulation

2. Applicability

This pathway applies to adult patients within Armadale Emergency Department with a suspected or confirmed lower limb DVT

3. Guidelines

This document is based on the [Diagnosis and Management of Venous Thromboembolism Guidelines \(2019\)](#) from the Thrombosis and Haemostasis Society of Australia and New Zealand, which aims to promote optimal management of VTE with an evidence-based approach (guideline summary available on [WILEY Online Library](#))

3.1 Determine the pre-test probability of DVT using Well's Criteria (two-level)

Well's DVT Criteria should only be applied to those patients who have been deemed at risk for DVT (based on history and clinical examination)

- If there is no concern for DVT, then there is no need for risk stratification

Clinical Feature	Score
Active cancer (treatment or palliation within 6 months)	1
Paralysis, paresis or recent plaster immobilisation of the lower extremity	1
Recently bedridden > 3 days, or major surgery within last 12 weeks	1
Localized tenderness along the distribution of the deep venous system	1
Entire lower limb swollen	1
Calf swelling > 3cm compared to asymptomatic leg (measured 10cm below tibial tuberosity)	1
Pitting oedema (confined to symptomatic leg)	1
Collateral (non-varicose) superficial veins present	1
Previously documented DVT	1
Alternative diagnosis as likely or more likely than that of DVT	-2
Score: 2 or more = DVT <u>likely</u> 1 or less = DVT <u>unlikely</u>	

3.2 Investigation for suspected DVT based on pre-test probability

- See [Diagnostic Imaging Pathways - Deep Venous Thrombosis \(Leg\)](#)
- If **DVT unlikely** on pre-test probability, request a D-dimer:
 - If age-adjusted D-dimer is negative, DVT is ruled out without imaging
 - If age-adjusted D-dimer is positive, proceed to imaging (Duplex ultrasound)
 - A positive D-dimer alone is not diagnostic of DVT

Age-adjusted D-dimer thresholds in patients older than 50 years

- Positive result = D-dimer >0.5mg/L
- If > 50 years old, positive result = D-dimer > (**age x 0.01mg/L**)
- Example: 60 years old, positive result = D-dimer > (60 x 0.01mg/L) = D-dimer >0.6mg/L

- Proceed direct to ultrasound, without requesting a D-dimer, if:
 - **DVT likely on pre-test probability**
 - Active cancer (<6 months since therapy/palliative stage)
 - DIC
 - Obvious infection
 - Inpatient (age >50 years)
 - Recent trauma or surgery <2 weeks previously
 - Third trimester of pregnancy
 - Symptoms >7days
- Request “Duplex ultrasound (entire lower leg)”, noting which leg to be scanned
 - Highly sensitive for proximal lower limb deep vein thrombosis
 - Less sensitive for deep calf vein thrombosis and for iliac vein thrombosis
- If ultrasound unable to be performed on the same day:
 - Therapeutic dose(s) LMWH until ultrasound available the next day
 - If [high bleeding risk](#) to discuss with senior
- If high clinical suspicion of DVT but ultrasound negative, or below knee component unable to be performed for technical reasons, perform a D-dimer
 - D-dimer negative: no further investigation for DVT required
 - D-dimer positive: repeat proximal lower limb ultrasound in 1 week (to exclude proximal propagation of calf thrombosis)

3.3 Determine if proximal or distal (below knee) DVT from ultrasound report

See [Major Deep and Superficial Veins of the Lower Limb](#)

- Proximal = thrombus present in popliteal vein or more proximal vein
- Distal = thrombus only occurs in the tibial, peroneal, gastrocnemius and/or soleal veins

3.4 Determine if DVT provoked or unprovoked

- Has impact on [risk of venous thromboembolism \(VTE\) recurrence](#) and [duration of anticoagulation](#)
- [Provoking factors](#) can be classified as:
 - Surgical (recent major surgery) vs non-surgical
 - Transient vs persistent
- The lifetime risk of VTE is 8%

3.5 Additional investigation for proven DVT

- All patients require assessment of contraindications to treatment and baseline bloods:
 - [Assess bleeding risk](#)
 - FBC/U+E/LFTs/Coags
 - BHCG for women of child-bearing age (if not already performed)
- If massive/unprovoked or recurrent DVT:
 - Malignancy screen (up to 10% of patients with unprovoked VTE are diagnosed with cancer in the year after VTE diagnosis)
 - History/examination
 - FBC/Ca⁺⁺/LFT/Urinalysis/CXR
 - Ensure age/sex appropriate cancer screening up to date
 - Mammogram/CST/prostate + PSA/FOB
 - Refer to GP to arrange
- A routine Thrombophilia screen is not indicated

3.6 Treatment options for confirmed DVT

- Depends on the [anatomical extent](#) of thrombus as well as patient factors
- Anticoagulation is indicated in most cases of DVT because it is highly effective in preventing thrombus extension or recurrence by at least 80%

ANTICOAGULATION

- [Medication options](#)
- [Duration of treatment](#)
- Discuss with Haematology Registrar (RPH) if uncertain or need more information
- Pregnant patients should be discussed with both Haematology and their Obstetrician

- Direct oral anticoagulants (DOACs) and warfarin are equally effective
- DOACs ([prescribing guidelines](#)):
 - Apixaban and Rivaroxaban are 1st line
 - Dabigatran is not yet listed on PBS for treatment of DVT
 - Do not require routine monitoring
 - Have very few drug/food interactions
 - Should not be used in pregnant or breastfeeding patients

- No dose adjustment required for extremes of body weight when treating DVT

3.7 Treatment options for confirmed Superficial Vein Thrombophlebitis (SVT)

- Self-limiting disorder but may be complicated by DVT or PE (particularly if [risk factors for extension](#))
- Antibiotics are not indicated unless evidence of sepsis
- [Information on SVT](#)
- [Treatment Algorithm](#)

Location/ Length	Treatment
≤ 3cm from saphenofemoral junction	Treat as per DVT (Therapeutic Dosing)
>3cm from saphenofemoral junction <u>AND</u> >5cm in length	Prophylactic Dosing for 45 days: Enoxaparin 40mg SC once daily (20mg if CrCl <30ml/min) or Rivaroxaban 10mg PO once daily (non-PBS)
>3cm from saphenofemoral junction <u>AND</u> <5cm in length	Topical or Oral NSAIDS for 7 days If severe symptoms or risk factors for extension, consider prophylactic anticoagulation as above

3.8 Risk Factors for Clot Extension

- Positive D-dimer
- Clot >5cm in length
- Involves multiple veins
- Unprovoked
- Malignancy
- Known thrombophilic disorder
- History of VTE
- HRT or pregnancy
- Recent surgery or trauma
- Inpatient admission
- Ongoing immobilisation

3.9 Post-Thrombotic Syndrome (PTS)

- Characterised by clinical features (e.g., swelling, discomfort, hyperpigmentation and lipodermatosclerosis) in a limb with previous DVT
- Occurs in approximately one in three patients following DVT
- Main risk factors for the development of PTS include:
 - DVT location (iliofemoral DVT increased risk compared to distal DVT)
 - Ipsilateral recurrence of DVT

- Prevention of PTS
 - Anticoagulation does not prevent PTS
 - If extensive DVT, thrombolysis can relieve venous outflow obstruction
 - Elastic Compression Stockings (ECS)
 - Conflicting findings in clinical trials for reduction in incidence of PTS
 - May be useful only to control symptoms of leg swelling and pain
 - GP can consider below knee graduated compression stockings

3.10 Disposition/Referral

- Massive, Iliofemoral or IVC DVT's require discussion with Vascular Team (RPH) and usually admission
- Most other patients can be discharged if adequate social support available and no other issues requiring admission
- [Referral for follow up](#) depends on patient factors and type of DVT

3.11 Provide with appropriate Patient Information Leaflets/Discharge Summary

- [WA Health Emergency Department Discharge Instructions - DVT](#)
- Apixaban/Rivaroxaban High-Risk Medication Patient Information Leaflets (CEC)
 - Under [Resources for Patients](#) tab at bottom of webpage
 - Translation available in English, Arabic, Chinese, Greek, Korean, Vietnamese
- Apixaban use in VTE
 - [Translated information: Arabic, Chinese, Greek, Vietnamese](#)
- Warfarin:
 - [Living with Warfarin](#)
 - [Translated information: Arabic, Italian, Khmer, Spanish, Vietnamese](#)

DISCHARGE SUMMARY to GP (copy to patient) documenting:

- DVT location (proximal/distal etc), and whether provoked/unprovoked
- Anticoagulation choice and expected duration of treatment
- Referral/follow up plan
- GP to consider graduated compression stockings if appropriate

4. Policy context

- HDWA
 - [High Risk Medication Management Policy](#)
 - [WA Anticoagulation Medication Chart](#)

5. Supporting information

The following documents support the implementation of this policy:

- [Deep Venous Thrombosis \(Leg, Suspected\)](#)
- [WILEY Online Library](#)

- [WA Health Emergency Department Discharge Instructions - DVT](#)
- [Venous Thromboembolism Prevention Guideline - Adult](#)
- [RPMG Clinical Practice Standard - VTE Risk Assessment and Prophylaxis for Adult Inpatients](#)

6. Compliance, monitoring and evaluation

The Safety, Quality, Education and Innovation Department can assist with determining monitoring requirements. Compliance monitoring should include the position and/or committees responsible for ensuring compliance, monitoring of KPI's, planned audits etc.

7. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 1 - Action 1.15 Identifies diversity and high-risk groups
- Standard 1 – Action 1.27 Evidence-based care
- Standard 4 – Action 4.11 Information for patients
- Standard 4 – Action 4.13 Information and decision support tools for medicine
- Standard 4 – Action 4.15 High risk medicines
- Standard 6 – Action 6.03 Partnering with consumers

8. References

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2. National Institute for Health and Care Excellence: Venous thromboembolic diseases: diagnosis, management and thrombophilia testing, NICE guideline [NG158], Published: 26 March 2020, updated 02 August 2023. Available from: <https://www.nice.org.uk/guidance/ng158> (accessed 07/01/2026)
3. Clinical Excellence Commission: Direct Oral Anticoagulants (DOAC) Guidelines, Version 1.1, March 2025. Available from: https://www.cec.health.nsw.gov.au/data/assets/pdf_file/0005/917807/DOAC-guidelines.pdf (accessed 07/01/2026)
4. BMJ Best Practice: Superficial Vein Thrombophlebitis (reviewed 07 December 2025) Available from: <https://bestpractice.bmj.com/topics/en-gb/335> (accessed 07/01/26)

9. Document owner

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10. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V3	10 02 2026	10 02 2029	Full review with only minor changes.

11. Authorisation

This guideline has been authorised and issued by the [enter position title of the authoriser].

Authorised by	Dr Tania Strickland - Director Clinical Services
Authorisation date	09 02 2026
Published date	10 02 2026

12. Appendices

12.1 Appendix1: Assessment of Bleeding Risk – HAS-BLED score

- See Table below or [MDCALC](#)
 - Risk stratification score to estimate the 1-year risk of major bleeding
- Bleed risk stratification scores should not be used to withhold treatment
 - Can be used to identify patients for monitoring or modified therapeutic approaches
 - Identification of some modifiable risk factors (e.g., blood pressure, medication use)
- Additional high-risk factors for bleeding:
 - Recent surgery / trauma (discuss with surgical team)
 - Active GI disease
 - Inherited or acquired bleeding disorder
 - High falls risk

	CONDITION	SCORE
H	Hypertension <i>Uncontrolled, SBP > 160mmHg</i>	1
A	Abnormal renal or liver function (1 point each) <i>Dialysis, transplant, Cr > 200</i> <i>Cirrhosis, Bilirubin > 2x normal with AST/ALT/AP > 3x normal</i>	1 or 2
S	Stroke <i>Prior history of stroke</i>	1
B	Bleeding history or pre-disposition	1
L	Labile INR <i>Unstable/High INRs, time in therapeutic range < 60%</i>	1
E	Elderly <i>Age > 65 years</i>	1
D	Drugs or Alcohol (1 point each) <i>Medications that may predispose to bleeding e.g. NSAIDS, antiplatelet agents</i> <i>Alcohol > 8 Std drinks/week</i>	1 or 2

Score 0 - 1 = Low risk

Score 2 = Moderate risk

Score ≥ 3 = High risk

12.2 Appendix 2: Major Deep and Superficial Veins of the Lower Limb

Deep Proximal	Deep Distal	Superficial
common iliac vein external iliac vein internal iliac vein common femoral vein profunda femoris vein (deep femoral vein) femoral vein* popliteal vein	gastrocnemius vein soleal vein tibial veins (anterior, posterior) peroneal veins	great saphenous vein small saphenous vein anterior accessory saphenous vein intersaphenous vein

*Note: the femoral vein was previously termed “superficial femoral vein”, but is actually a deep vein

12.3 Appendix 3: Provoking risk factors (non-surgical)

Type of Risk Factor	Condition
Transient	• Acute medical illness with immobilisation for at least 3 days • Oestrogen therapy • Pregnancy/post-partum • Leg injury associated with reduced mobility for at least 3 days • Long-haul travel
Persistent	• Active cancer • Antiphospholipid syndrome • Ongoing non-malignant condition associated with a twofold or higher increased risk of recurrent VTE after stopping anticoagulant therapy (i.e., inflammatory bowel disease and other chronic inflammatory states)

12.4 Appendix 4: Risk of VTE recurrence

Strong Risk Factors for Recurrence	
	Unprovoked VTE
	Prior VTE
	PE or proximal DVT
	Persistent risk factor (e.g. active cancer, antiphospholipid syndrome)
	Antithrombin, protein C & S deficiency
Moderate Risk Factors for Recurrence	
	VTE provoked by non-surgical risk factor
	Male sex
	Elevated D-dimer after cessation of anticoagulation
Factors that have little or no effect on recurrence	
	Factor V Leiden or prothrombin gene heterozygosity
	Residual thrombus on imaging

Type of VTE	Recurrence Rate at one Year after Stopping Anticoagulation	Recurrence Rate at five Years after Stopping Anticoagulation
First VTE provoked by major surgery or major trauma	1%	3%
First VTE provoked by transient risk factor (non-surgical)	5%	15%
Provoked VTE with persistent risk factors e.g., active cancer	15%	45%
First unprovoked distal DVT	5%	15%
First unprovoked proximal DVT	10%	30%
Second episode of unprovoked VTE	15%	45%

12.5 Appendix 5: Anticoagulation options

Medication	Initiation (ED to Prescribe)	Maintenance (GP to Prescribe)	Indication	Contraindications
RIVAROXABAN (Xarelto)	15mg BD for 3 weeks Authority required (Streamlined): 4098	20mg OD		CrCl < 30ml/min* Active bleeding Significant bleeding disorder Severe hepatic impairment Pregnancy/Breastfeeding
APIXABAN (Eliquis)	10mg BD for 7 days Authority required (Streamlined): 4098	5mg BD		CrCl < 25ml/min* Active bleeding Significant bleeding disorder Severe hepatic impairment Pregnancy/Breastfeeding
LMWH	Enoxaparin 1.5mg/kg daily or 1mg/kg BD (TBW#) 1mg/kg daily if CrCl < 30ml/min* 1mg/kg BD if BMI ≥30kg/m ²		Pregnancy Breastfeeding Oncology Patient Use until Warfarin level is therapeutic	HITS Discuss with Haematology for dose adjustment and monitoring advice if: CrCl < 30ml/min* BMI ≥30kg/m²
Warfarin	Refer to WA Anticoagulation Chart Start Warfarin and LMWH simultaneously Continue LMWH for at least 5 days and until INR ≥ 2	Aim INR 2.0 – 3.0	1 st line in Anti-Phospholipid Syndrome or Lupus Anticoagulant DOACS contraindicated	Pregnancy Allergy

Total Body Weight

*CrCl calculated using [Cockcroft-Gault equation](#), not eGFR reported on biochemistry results

12.6 Appendix 6: Duration of Anticoagulation

Thrombus Type	Anticoagulation Duration	Disposition / Referral
Massive DVT – Iliofemoral/IVC	3 months minimum (duration determined by Haematology)	ADMIT D/w Vascular Team, RPH
Any DVT provoked by active cancer	At least 6 months therapeutic LMWH	Discharge* Refer own Oncology Team
Proximal DVT (unprovoked or transient non-surgical risk factor)	3 – 6 months (duration determined by Haematology)	Discharge* eReferral to Haematology, RPH
Proximal DVT (provoked by major surgery/trauma that is no longer present)	3 months	Discharge* GP follow up
Distal DVT (unprovoked or persistent risk factor)	3 months	Discharge* 1 st DVT - GP follow up Recurrent DVT - eReferral to Haematology, RPH
Distal DVT (transient risk factor resolved)	6 weeks	Discharge* GP follow up
Superficial Thrombophlebitis (≥ 5cm length and/or within 3cm of SFJ or SPJ)	45 days Treatment Algorithm	Discharge* GP follow up

*if adequate social support present